

Relations and Functions: Formula Sheet

Class 12 Mathematics – Chapter 1 Complete Reference & Formulas

1 Types of Relations on a Set A

- **General Relation (R):** A subset of the Cartesian product $A \times A$, i.e., $R \subseteq A \times A$.

- **1. Empty Relation:**

$$R = \phi \subset A \times A$$

No element of A is related to any element of A .

- **2. Universal Relation:**

$$R = A \times A$$

Each element of A is related to every element of A . (Empty and universal relations are called *trivial relations*).

- **3. Reflexive Relation:**

$$(a, a) \in R, \quad \forall a \in A$$

- **4. Symmetric Relation:**

$$(a_1, a_2) \in R \implies (a_2, a_1) \in R, \quad \forall a_1, a_2 \in A$$

- **5. Transitive Relation:**

$$(a_1, a_2) \in R \text{ and } (a_2, a_3) \in R \implies (a_1, a_3) \in R, \quad \forall a_1, a_2, a_3 \in A$$

2 Equivalence Relations & Equivalence Classes

- **Equivalence Relation:** A relation R on set A that is simultaneously **reflexive**, **symmetric**, and **transitive**.
- **Equivalence Class $[a]$:** For an equivalence relation R on set X , the equivalence class containing $a \in X$ is defined as:

$$[a] = \{x \in X : (x, a) \in R\}$$

Fundamental Properties of Equivalence Classes:

- All elements in $[a]$ are related to each other.
 - Two equivalence classes are either identical or completely disjoint: $[a] \cap [b] = \phi$ if $[a] \neq [b]$.
 - The union of all pairwise disjoint equivalence classes forms the whole set X : $\bigcup [a] = X$.
- **Intersection of Equivalence Relations:** If R_1 and R_2 are two equivalence relations on set A , then $R_1 \cap R_2$ is also an equivalence relation.

3 Types of Functions (Mappings)

- **1. One-One (Injective) Function:**

$$f(x_1) = f(x_2) \implies x_1 = x_2, \quad \forall x_1, x_2 \in X$$

Equivalently: $x_1 \neq x_2 \implies f(x_1) \neq f(x_2)$. (A function that is not one-one is called **many-one**).

- **2. Onto (Surjective) Function:**

$$\forall y \in Y, \exists x \in X \text{ such that } f(x) = y$$

Equivalently: $\text{Range}(f) = \text{Co-domain}(Y)$.

- **3. Bijective Function:** A function $f : X \rightarrow Y$ that is both **one-one (injective)** and **onto (surjective)**.

4 Composition of Functions & Invertibility

- **Composition of Functions ($g \circ f$):** Let $f : A \rightarrow B$ and $g : B \rightarrow C$. The composite function $g \circ f : A \rightarrow C$ is defined as:

$$(g \circ f)(x) = g(f(x)), \quad \forall x \in A$$

Note: In general, $g \circ f \neq f \circ g$.

- **Invertible Function & Inverse (f^{-1}):** A function $f : X \rightarrow Y$ is invertible if there exists $g : Y \rightarrow X$ such that:

$$g \circ f = I_X \quad \text{and} \quad f \circ g = I_Y$$

Where I_X and I_Y are identity functions on X and Y respectively ($I_X(x) = x$).

$$f \text{ is invertible} \iff f \text{ is bijective (one-one and onto)}$$

5 Counting Theorems & Finite Set Properties

- **Finite Set Bijectivity Theorem:** Let X be a finite set. A function $f : X \rightarrow X$ is one-one if and only if f is onto.
- **Total Number of Functions from Set A to Set B :** If $n(A) = m$ and $n(B) = n$:

$$\text{Total Functions} = n^m$$

- **Total Number of One-One (Injective) Functions ($f : A \rightarrow B$):**

$$\text{Total One-One Maps} = \begin{cases} {}^nP_m = \frac{n!}{(n-m)!}, & \text{if } n \geq m \\ 0, & \text{if } n < m \end{cases}$$

- **Total Number of Bijective / Onto Functions on Finite Set A ($n(A) = n$):**

$$\text{Total Bijections} = n!$$